Differentiating Reciprocal and Inverse Trig Functions

Exam code: 9MA0

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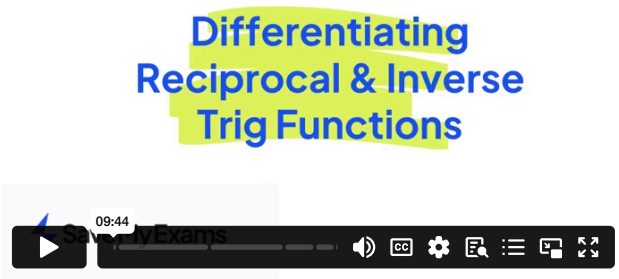

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Help me understand

Differentiating reciprocal and inverse trig functions

How do I differentiate reciprocal trigonometric functions?

- The formulae for the derivatives of the reciprocal trigonometric functions are:

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$

$$\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

- You can derive the derivatives for **sec**, **cosec**, and **cot** using the **chain rule** and the derivatives of the basic trigonometric functions

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

$$\frac{dy}{dx} = -(\cos x)^{-2}(-\sin x)$$

USE CHAIN RULE FOR $(f(x))^n$

$n(f(x))^{n-1}$

$f'(x)$

$$\frac{dy}{dx} = \frac{\sin x}{\cos^2 x} = \left(\frac{1}{\cos x}\right)\left(\frac{\sin x}{\cos x}\right)$$

$$\frac{dy}{dx} = \sec x \tan x$$

$\tan x = \frac{\sin x}{\cos x}$

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How do I differentiate inverse trigonometric functions?

- The formulae for the derivatives of the inverse trigonometric functions are:

$$\frac{d}{dx}(\arcsin x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\arccos x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\arctan x) = \frac{1}{1+x^2}$$

- You can derive the derivatives of **arcsin**, **arccos**, and **arctan** using the reciprocal form of the **chain rule** and the derivatives of the basic trigonometric functions

e.g. DIFFERENTIATE $y = \arccos x$

IF $y = \arccos x$
THEN $x = \cos y$

arccos AND cos ARE INVERSES

$\frac{dx}{dy} = -\sin y$

DIFFERENTIATE WITH RESPECT TO y

BUT $\sin^2 y + \cos^2 y = 1$
SO $\sin y = \sqrt{1-\cos^2 y}$

USE TRIG IDENTITY

AND $\cos y = x$
SO $\sin y = \sqrt{1-x^2}$

$\frac{dx}{dy} = -\sin y = -\sqrt{1-x^2}$

AND $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$

USE RECIPROCAL FORM OF CHAIN RULE

SO $\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$

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Examiner Tips and Tricks

- The formulae for the derivatives of **sec x**, **cosec x**, and **cot x** are in the formulae booklet – you don't need to memorise them

arctan are not in the formulae booklet

- You should know how to derive those derivatives using the reciprocal form of the chain rule

Worked Example

a) Given that $y = 5\text{cosec}^2 x$ find $\frac{dy}{dx}$

b) Show that the derivative of $\arctan x$ is $\frac{1}{1+x^2}$

a) $y = 5(\text{cosec}\, x)^2$ ← $5(f(x))^2$ — USE CHAIN RULE

$\frac{dy}{dx} = 10\text{cosec}\, x\,(-\text{cosec}\, x \cot x)$ ← $5 \times 2(f(x))^{2-1} f'(x)$

$\frac{dy}{dx} = -10\text{cosec}^2 x \cot x$

b) LET $y = \arctan x$

THEN $x = \tan y$ ← tan AND arctan ARE INVERSES

SO $\frac{dx}{dy} = \sec^2 y$ ← DIFFERENTIATE WITH RESPECT TO y

$\frac{dx}{dy} = 1 + \tan^2 y$ ← USE TRIG IDENTITY

$\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = \frac{1}{1+\tan^2 y}$ ← RECIPROCAL FORM OF CHAIN RULE

BUT $x = \tan y$ SO

$\frac{dy}{dx} = \frac{1}{1+x^2}$

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